CHEMICAL REACTIONS 14

Chemistry is a science that studies the composition and properties of substances and how those substances interact and transform to create new substances. **Physical changes** involve the changing of a substance from one state to another without altering the chemical nature of the substance. For example, melting ice and boiling water are physical changes. The bonds between the water molecules are affected but the molecules remain unchanged. In a **physical change**, no new substance is formed and it is relatively easy to reverse the change.

1 What physical change is the 'opposite' of boiling? _____

2 Why is electricity passing through a light globe a physical change?

_____

3 Give an example of a physical change not already mentioned. _____

Chemical reactions occur when atoms are rearranged to form new substances (compounds). In a chemical change, new substances are formed when atoms form new bonds with each other. Chemical changes are called **chemical reactions**. Chemical reactions happen when one or more of the following take place:

- A change in temperature
- A new substance appears
- The original substance disappears
- A change is difficult to reverse.

4 State whether the following are physical changes or chemical changes.

a Soup from a can is warmed in a saucepan. _____

b Ice-cream melts. _____

c A tomato ripens. _____

d Sugar dissolves in a cup of coffee. _____

e Milk turns sour. _____

f Water is added to a glass of cordial. _____

g A mixture of flour, water and yeast is heated to make bread. _____

5 Name another chemical reaction not already mentioned. _____

Energy is often needed to start a chemical reaction. A match must be struck against the side of a matchbox to ignite it. A naked flame must be placed next to gas coming from a Bunsen burner to light it. These are examples of **activation energy**. Not all chemical reactions occur at the same rate (speed). Magnesium (Mg) and gold (Au) are both shiny metals. A piece of magnesium can burn in a few seconds but gold does not burn – it may take centuries for a gold coin to even tarnish (go dull).

6 Explain why cooking an egg is classified as a chemical reaction.

_____

7 What type of energy is needed to charge a car battery? _____

8 What types of energy are given out by a lit Bunsen burner? _____

9 Give a new example of a very fast chemical reaction. _____

10 Give a new example of a very slow chemical reaction. _____

11 What is *activation energy*? _____

12 When a candle burns, both physical changes and chemical reactions occur.

a Describe a chemical change when a candle burns. _____

b Describe a physical change when a candle burns. _____

13 Explain why chemical changes are harder to reverse than physical changes.

_____

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In a chemical reaction, the substances are divided into two groups: **reactants** and **products**. The reactants are the materials you begin with and they react with each other. The products are the new substances produced during the reaction. Reactions are not all the same; the reactants and products can be a mixture of solids, liquids and gases. Some reactions give out heat (are **exothermic**). Other reactions take in heat from their surroundings (are **endothermic**).

14 In a reaction, what is meant by:

a reactants? _____

b products? _____

As a reaction proceeds, the concentration of reactants decreases and the concentration of products increases. The rate of reaction can be calculated by measuring the change in either reactants or products against time. This can be expressed graphically, as shown in figures 1 and 2. The x-axis is time and the y-axis is concentration.

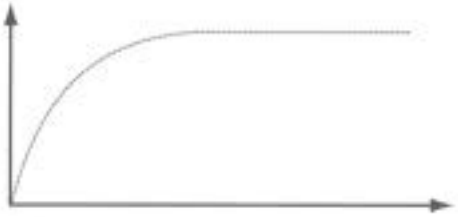


Figure 1



Figure 2

15 Describe what is happening in:

a Figure 1 (products). _____

b Figure 2 (reactants). _____

16 Explain what is meant by the following terms.

a Exothermic _____

b Endothermic _____

Three common types of reactions are **synthesis**, **combustion** and **decomposition**. Synthesis involves the formation of a compound from the elements that make it up. Combustion is sometimes described as burning; an element combines rapidly with oxygen, gives out heat and forms an oxide. In decomposition, a compound is broken down into simpler substances.

17 For each reaction, write a word equation and classify it as synthesis, combustion or decomposition.

a $2\ HgO\ (s) \rightarrow 2\ Hg\ (l) + O_2\ (g)$

_____

b $H_2\ (g) + Br_2\ (g) \rightarrow 2\ HBr\ (g)$

_____

c $CH_4\ (g) + 2O_2\ (g) \rightarrow CO_2\ (g) + 2H_2O\ (l)$ (CH_4 = methane)

_____

d $N_2\ (g) + 3\ H_2\ (g) \rightarrow 2\ NH_3\ (g)$ (NH_3 = ammonia)

_____

e $CaCO_3\ (s) \rightarrow CaO\ (s) + CO_2\ (g)$

_____

18 Give an everyday example of synthesis, combustion and decomposition.

19 In a chemical reaction, the number of atoms in the reactants equals the number of atoms in the products. True or false?

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